ANALYSIS OF STRUCTURES

Analysis of Planar Trusses using the Method Of Joints.
Analysis of Planar Trusses using the Method Of Sections.
Analysis of Frames.

ANALYSIS OF STRUCTURES

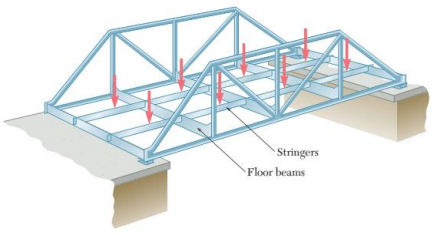
- Structural engineering is the field of engineering mostly associated with the design and analysis of engineering structures.
- Engineering structures basically, are designed to support, transmit or modify forces safely. They normally comprise of a number of members connected to form a main structure.
- There are different types of engineering structures, but focus will be on trusses and frames.
- Trusses are designed to transmit forces over long distances and comprise straight, **slender** bars that are joined together to form in most cases a pattern of triangles.
- The loads on trusses are always applied at the joints, while in frames, loads may not necessarily be applied at the joints. Also, members of trusses are normally two force members, whereas in frames, at least one member is a multiforce member.

ANALYSIS OF STRUCTURES
Trusses

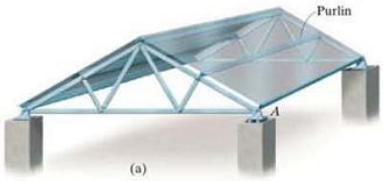
- Trusses may be classified as simple or compound.
- A truss is therefore often considered simple if its design or construction is such that each time we add 2 new members, they are joined such the number of joints in the truss increases by 1.
- A Truss is designed such that its members are two are two-force members (i.e. they are either in tension or compression).
- In designing a truss, it is desirable to know the force each individual member must sustain.

ANALYSIS OF STRUCTURES
Trusses

- Common applications are roofs, bridges and power pylons.



Bridge Truss
Source: Mechanics for Engineers by Beet *et al*



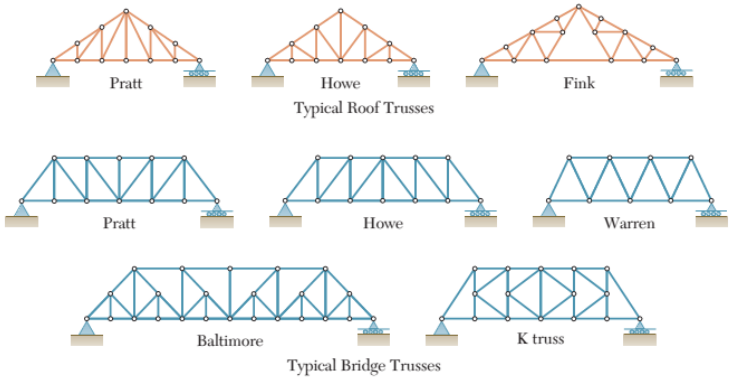
Roof Supporting Truss
Source: Engineering Mechanics Statics by Hiebbler



Power Pylon
Source: http://upload.wikimedia.org/wikipedia/commons/7/7e/Electricity_pylon_power_outage.jpg

ANALYSIS OF STRUCTURES
Planar Trusses

- Trusses may be Planar or Space Trusses.



Some Planar Trusses
Source: Beet *et al*

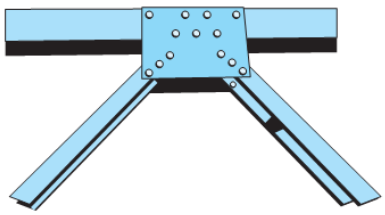


Space Truss
Source
*http://www.picstopin.com/1000/truss/http:%7C%7Cwww*timplex*com%7Ctrusstypess*png/*

ANALYSIS OF STRUCTURES
Analysis of Planar Trusses

- Assumptions for analysis:
  - The weights of the slender bars/members are negligible.
  - Forces act at the ends of the members such that they are in either tension or compression.
  - All joints are pins.

- There are two approaches to analysis:
  - The method of Joints
  - The method of Sections.



Truss members welded or riveted to a gusset plate
Source: Engineering Mechanics – Statics by Pytel

ANALYSIS OF STRUCTURES
Analysis of Planar Trusses - The Method of Joints

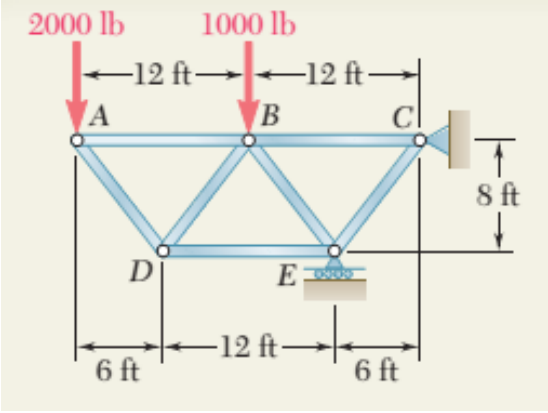
- This involves an equilibrium analysis on each joint to determine the forces being exerted on the end of each member at that joint.
- It is based on the assumption that if the whole truss is in equilibrium, each of its members or joints are also in equilibrium.

- Analysis is done in two main steps:
  - Determine the reactions at the support reactions using the FBD of the entire truss.
  - Conduct an equilibrium analysis at each joint/pin to determine the forces in each member.

ANALYSIS OF STRUCTURES
Analysis of Planar Trusses - The Method of Joints

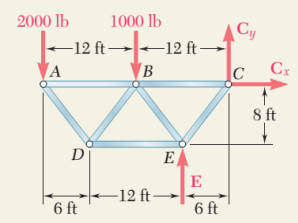
- Example

Determine the force in each member of the truss shown below using the method of joints.



ANALYSIS OF STRUCTURES
Analysis of Planar Trusses - The Method of Joints

- Solution
  - Equilibrium analysis on the entire truss to determine the support reactions



For equilibrium,

$$+\rightarrow \sum F_x = 0 : C_x = 0$$

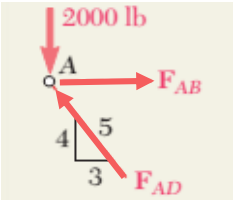
$$+\uparrow \sum F_y = 0 : C_y + E = 3000\ \text{lb}$$

$$\circlearrowleft \sum M_C = 0 : (6\ \text{ft})E - (2000\ \text{lb})(24\ \text{ft}) - (1000\ \text{lb})(12\ \text{ft}) = 0$$

$$E = 10000\ \text{lb}\uparrow$$

- Equilibrium analysis on the individual joints

Joint A



For equilibrium,

$$+\rightarrow \sum F_x = 0 : F_{AB} - \frac{3}{5}F_{AD} = 0 \qquad \text{----(1)}$$

$$+\uparrow \sum F_y = 0 : \frac{4}{5}F_{AD} - 2000\ \text{lb} = 0 \qquad \text{----(2)}$$

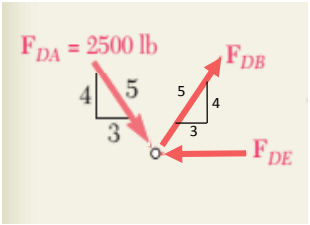
Solving (1) and (2) simultaneously,

$$F_{AD} = 2500\ \text{lb} \qquad F_{AB} = 1500\ \text{lb}$$

ANALYSIS OF STRUCTURES
Analysis of Planar Trusses - The Method of Joints

- Solution

Joint D



For equilibrium,

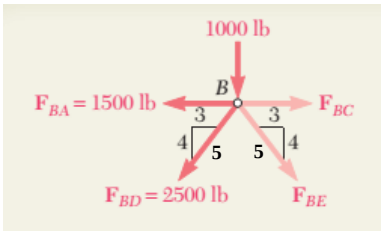
$$+\rightarrow \sum F_x = 0 : \frac{3}{5}(2500\ \text{lb}) - F_{DE} + \frac{3}{5}F_{DB} = 0 \qquad \text{----(1)}$$

$$+\uparrow \sum F_y = 0 : \frac{4}{5}F_{DB} - \frac{4}{5}(2000\ \text{lb}) = 0 \qquad \text{----(2)}$$

Solving (1) and (2) simultaneously,

$$F_{DB} = 2500\ \text{lb} \qquad F_{DE} = 3000\ \text{lb}$$

Joint B



For equilibrium,

$$+\rightarrow \sum F_x = 0 : -\frac{3}{5}(2500\ \text{lb}) - 1500\ \text{lb} + F_{BC} + \frac{3}{5}F_{BE} = 0 \qquad \text{----(1)}$$

$$+\uparrow \sum F_y = 0 : -1000\ \text{lb} - \frac{4}{5}(2000\ \text{lb}) - \frac{4}{5}F_{BE} = 0 \qquad \text{----(2)}$$

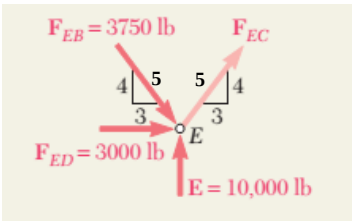
Solving (1) and (2) simultaneously,

$$F_{BE} = 3750\ \text{lb} \qquad F_{BC} = 5250\ \text{lb}$$

ANALYSIS OF STRUCTURES
Analysis of Planar Trusses - The Method of Joints

➢Solution

Joint E



For equilibrium,

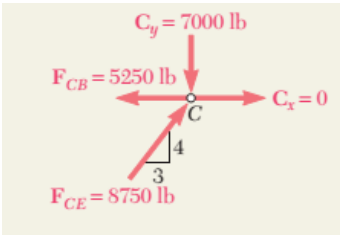
$$+\rightarrow \sum F_x = 0 : \frac{3}{5}(3750\,\text{lb}) + 3000\,\text{lb} + \frac{3}{5}F_{EC} = 0 \qquad \text{----(1)}$$

$$+\uparrow \sum F_y = 0 : -\frac{4}{5}(3750\,\text{lb}) + \frac{4}{5}F_{EC} + 10000\,\text{lb} = 0 \qquad \text{----(2)}$$

Solving (1) and (2) simultaneously,

$F_{DB} = 2500\,\text{lb}$ $\quad F_{DE} = 3000\,\text{lb}$

Joint C



For equilibrium,

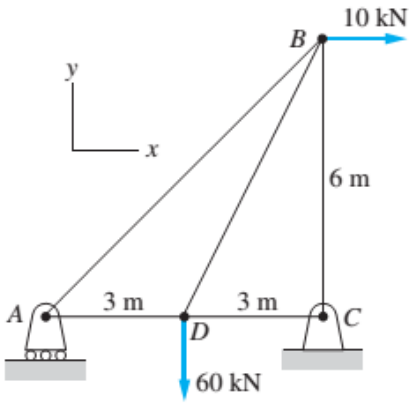
$$+\rightarrow \sum F_x = 0 : \frac{3}{5}(8750\,\text{lb}) - 5250\,\text{lb} + 0 = 0$$

$$+\uparrow \sum F_y = 0 : -7000\,\text{lb} - \frac{4}{5}(8750\,\text{lb}) = 0$$

ANALYSIS OF STRUCTURES
Analysis of Planar Trusses - The Method of Joints

➢Example

Determine the force in each member of the truss shown below using the method of joints. Indicate whether each member is in tension or compression



$P_{AB} = 28.3$ kN (Compression)

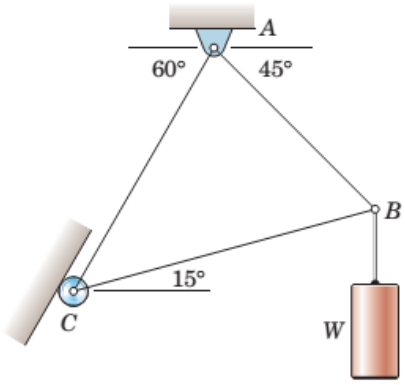
$P_{AD} = 20$ kN (Tension)

$P_{BC} = 40$ kN (Compression)

ANALYSIS OF STRUCTURES
Analysis of Planar Trusses - The Method of Joints

Question

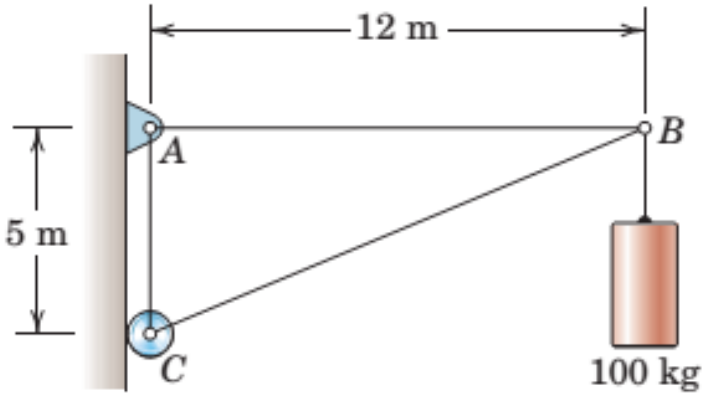
Determine the force in each of the members of the loaded truss



ANALYSIS OF STRUCTURES
Analysis of Planar Trusses - The Method of Joints

➢Question

Determine the force in each of the members of the loaded truss



ANALYSIS OF STRUCTURES
Analysis of Planar Trusses - The Method of Joints

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Determine the force in each of the members of the loaded truss

